

2024 Ph H2 Q6

Section: Particles and Waves

Topic: Electron beams; Electric & Magnetic Fields

Summary of Question

Electrons are thermionically emitted from a heated cathode and accelerated through 2.40 kV towards an anode 45.0 mm away. A small hole in the anode produces a narrow electron beam that then enters a uniform magnetic field between coils.

Tasks:

- (a)(i) Complete the electric field pattern between cathode (-) and anode (+).
- (a)(ii) Using $\Delta E_k = 3.84 \times 10^{-16} \text{ J}$, calculate the maximum speed of an electron at the anode.
- (a)(iii) Calculate the maximum acceleration of an electron between the electrodes (2.40 kV, 45.0 mm).
- (b) Determine the direction of the uniform magnetic field from the deflection of the beam.

(a)(i) Electric field pattern

Field lines start on the anode (+) and end on the cathode (-), directed from + to -, perpendicular to the plates, uniformly

spaced between the flat faces, curving near the edges.

Final answer: Draw straight, evenly spaced lines from anode to cathode (with arrows from + to -); label + on anode and - on cathode.

(a)(ii) Maximum speed at the anode

Relationship: $\Delta E_k = \frac{1}{2} m v^2 \Rightarrow v = \sqrt{(2\Delta E_k / m)}$.

Substitution: $v = \sqrt{(2 \times 3.84 \times 10^{-16} / 9.11 \times 10^{-31})} = 2.903 \times 10^7 \text{ m s}^{-1}$.

Final answer: $v \approx 2.90 \times 10^7 \text{ m s}^{-1} (\approx 29.0 \times 10^6 \text{ m s}^{-1})$.

(a)(iii) Maximum acceleration between cathode and anode

Uniform field between parallel electrodes: $E = V/d$. Electron acceleration: $a = eE/m = eV/(m d)$.

Substitution: $a = (1.60 \times 10^{-19} \times 2.40 \times 10^3) / (9.11 \times 10^{-31} \times 45.0 \times 10^{-3}) = 9.367 \times 10^{15} \text{ m s}^{-2}$.

Final answer: $a \approx 9.37 \times 10^{15} \text{ m s}^{-2}$.

(b) Direction of the uniform magnetic field

Using the right-hand rule for $v \times B$ (and reversing for negative charge), a beam that bends upward while moving to the right

implies B is into the page.

Final answer: Magnetic field into the page.

Revision Tips

Electron energy gain from a potential difference: $\Delta E_k = eV$ (per electron).

Between parallel plates, $E = V/d$ (neglecting edge effects); $a = eE/m$.

For charged particles in B fields, use the $\mathbf{v} \times \mathbf{B}$ rule and remember electrons ($-q$) deflect opposite to the right-hand direction.